EEE 4484

Lab Report : 03

Experiment Name: Study of Operational Amplifier as Triangular Wave Generator & Astable Multivibrator

Name: FARHAN SHAHRIYAR HOSSAIN

ID : 220041113

Section : 1A

Dept : CSE

Objective:

- Understand the fundamental functions and applications of an operational amplifier.
- Explore how to design and analyze a bipolar triangular wave generator using an op-amp.
- Learn the working principle and construction of a unipolar triangle wave generator with an op-amp.
- Develop and test an astable multivibrator circuit using an op-amp to generate continuous waveforms without external triggering.

Equipment:

- DC power supply
- Signal / Function Generator
- Bread board
- Resistors
- Diode (IN 4007)
- Capacitors
- OP-Amp (LM 741)
- Oscilloscope
- Connecting wires

Bipolar Triangle Wave Generator using Op-Amp:

Circuit Diagram:

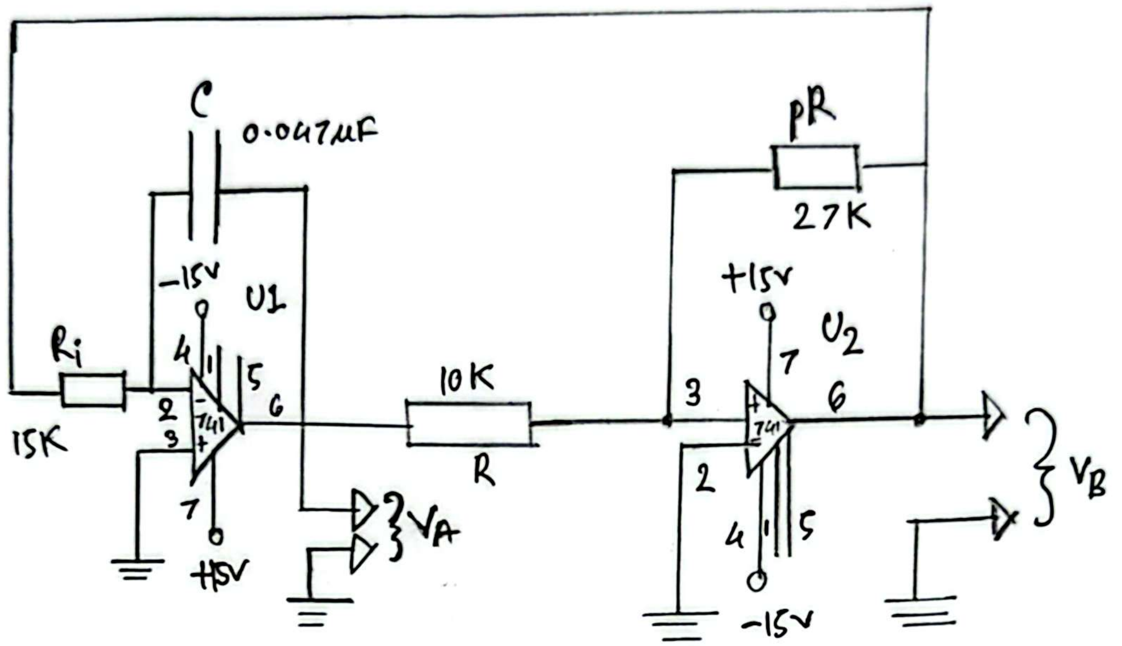


Fig: Bipolar Triangle wave Generator

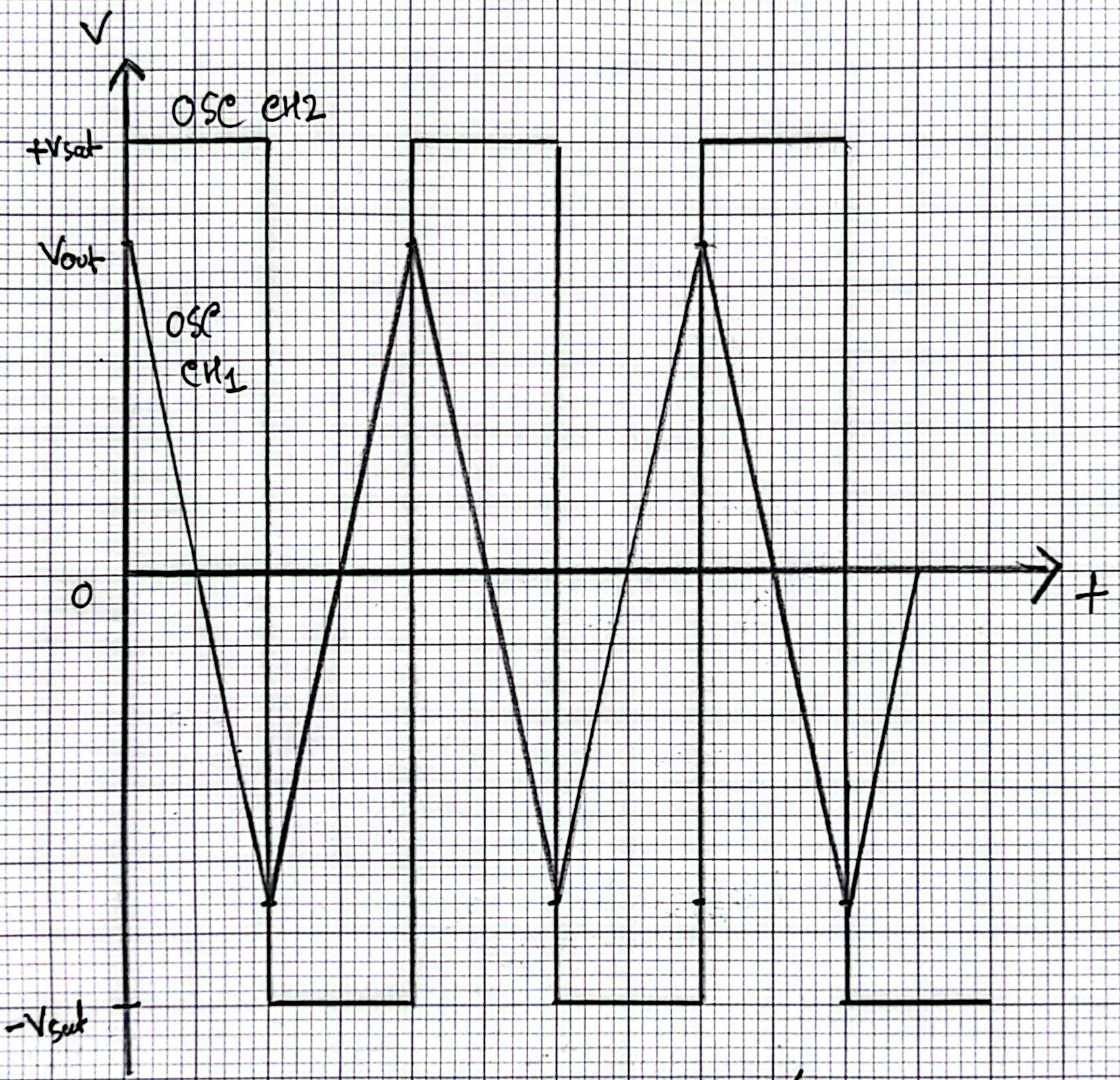
Justification:

The comparator, U2 switches its output between the positive and negative saturation voltages (+15V and -15V) depending on the input from the integrator U1. This output behaves like square wave and is fed back to the input of the integrator through resistor R. The integrator receives the square wave signal at its input & produces

2 mm GRAPH PAPER

Examining body		Centre number					
Candidate name		Candidate number					
Paper refernce		Question number		Sheet number			

Fig: Bipolar Triangle Wave Generator



OSC CH1 Pk-Pk 10.6 ✓

OSC CH2 Pk-Pk 27.6 ✓

a linearly increasing or decreasing voltage across the capacitor C, forming a triangular waveform. The output voltage is fed back into the non-inverting terminal of the comparator through resistor network R_i, pR.

Unipolar triangle Wave Generator using OP-Amp:

Circuit diagram:

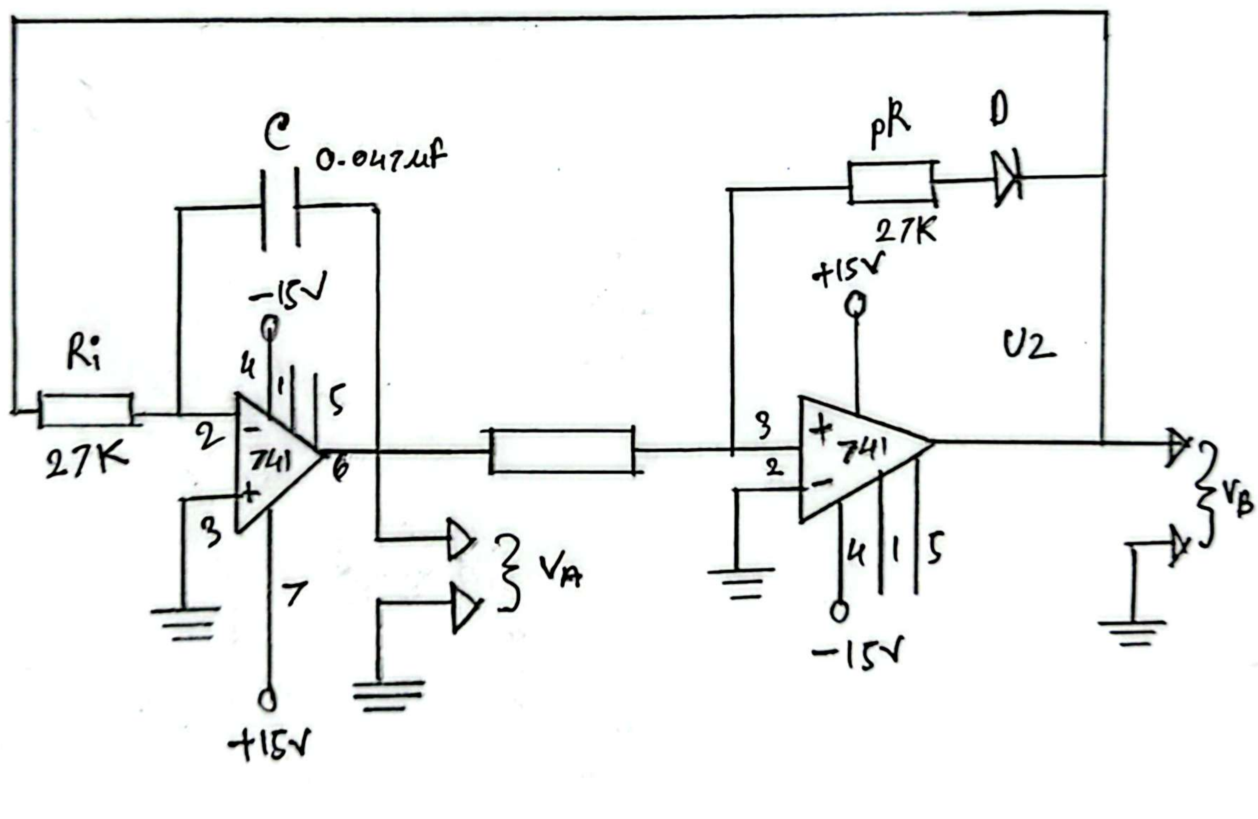
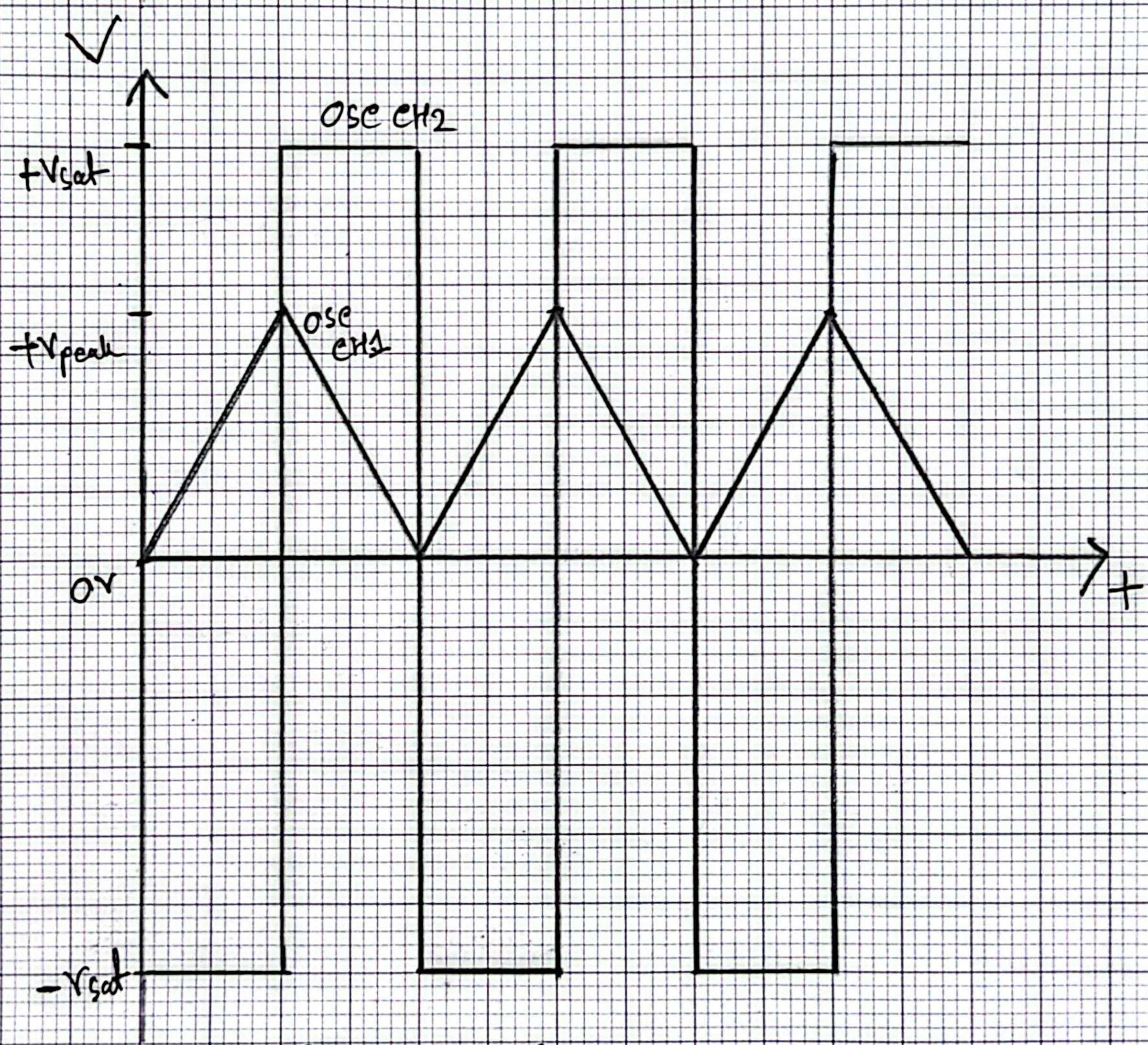


Fig: Unipolar Triangle Wave Generator

2 mm GRAPH PAPER

Examining body		Centre number							
Candidate name		Candidate number							
Paper refernce		Question number		Sheet number					

Fig: Unipolar Triangle Wave Generator



OSC CH1 Pk-Pk : 4.88V

OSC CH2 Pk-Pk : 28.0V

Justification:

Here, U2 operates as a comparator produceing a wave of square shape while U1 acts as an integrator to generate the triangular waveform. The diode in the feedback path ensured that the comparator output stayed within a fixed range instead of oscillating between ±V as in bipolar version. This effectively shifts the triangle waveform to the positive region, achieving unipolar behavior.

Astable Multivibrator using OP-Amp:

Circuit Diagram:

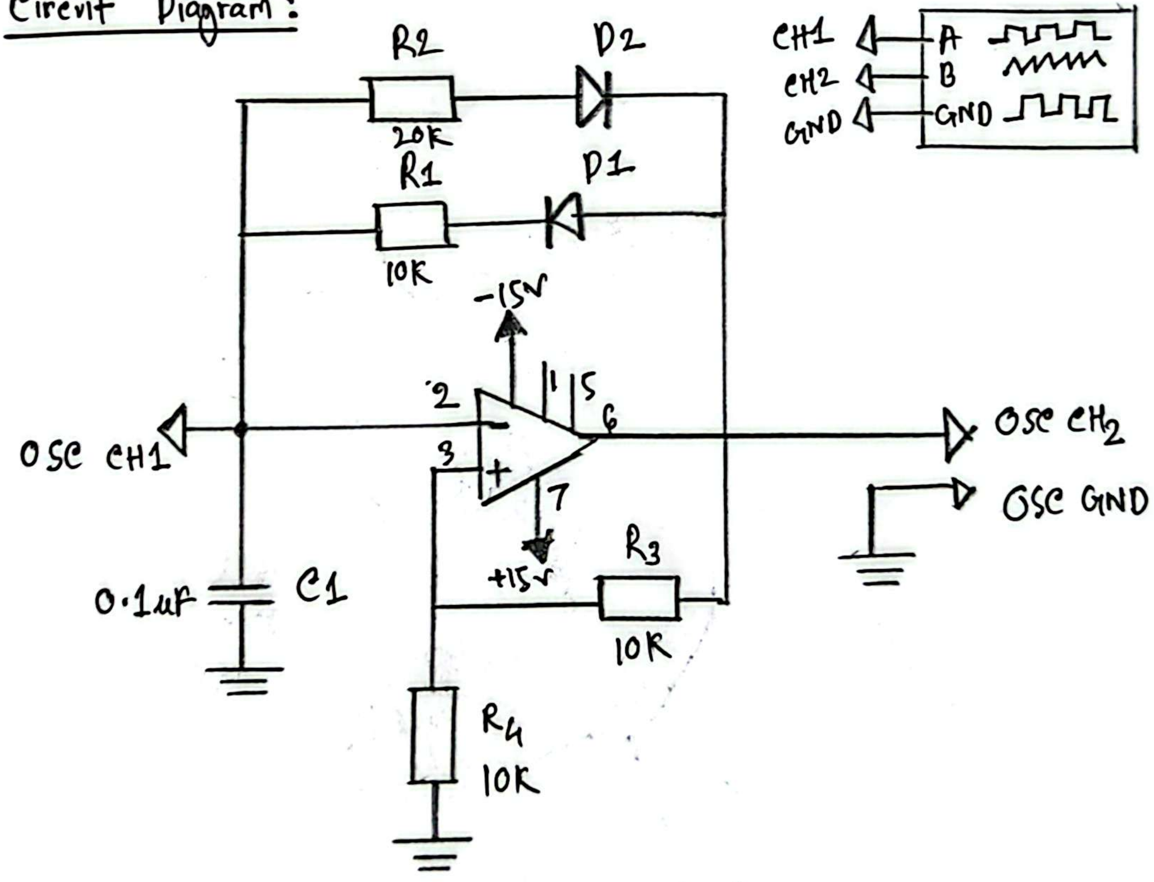
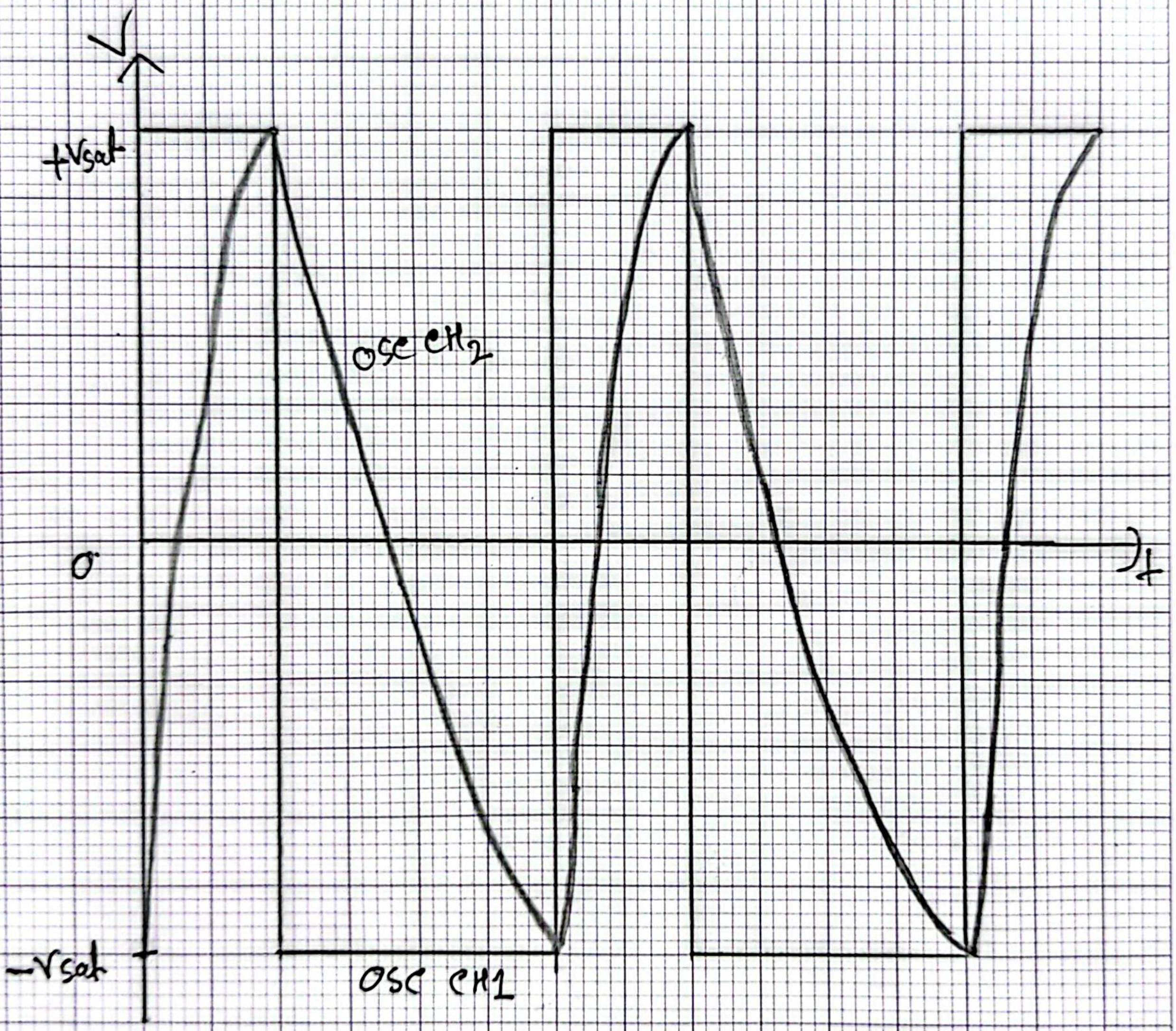


Fig: Astable Multivibrator

2 mm GRAPH PAPER

Examining body		Centre number	
Candidate name		Candidate number	
Paper refernce		Question number	Sheet number

Fig: Astable Multivibrator



OSC CH1 PK-PK : 28.8 V

OSC CH2 PK-PK : 14.2 V

Justification:

The circuit was configured with positive feedback and diode steering to generate a continuous square wave without any external triggering. The RC network (C1, R1, R2) determines the charging / discharging cycle of the capacitor, while diodes D1, D2 direct current flow to control timing during each half-cycle. As the capacitor charges and discharges, the OP-Amp output toggles between saturation levels producing an oscillating output signal. The waveform observed on the oscilloscope confirmed stable, periodic wave generation.

Answers to Report Questions:

Answer to Question 1 - Both circuits in Fig 3(c) and fig 3(d) are triangle wave generators having primary difference in the nature of output waveform:

> Fig 3(c) generates bipolar triangle wave, where the waveform oscillates symmetrically around 0V. The comparator output

swings between positive and negative saturation, feeding the integrator to produce a centered triangle wave.

> fig 3(d) produces a unipolar triangle wave, where the waveform remains entirely above 0V. This is achieved by adding a diode at the output of the comparator which clamps negative part of the waveform, allowing only positive voltages to reach the integrator. As a result, triangle wave shifts upwards and becomes unipolar.

Answer to Question 2 : The circuit in fig 3(e) is an astable vibrator. It functions as a self-oscillating square wave generator with no external input. The key components include a capacitor (C1), resistors (R1, R2, R3, R4) and diodes (D1, D2). The capacitor charges and discharges alternatively through R1, R2 while the OP-Amp provides positive feedback via voltage dividers and the diode network. When the capacitor voltage reaches the swinging threshold determined by the feedback path, the OP-Amp output toggles between high and low saturation levels. This transition reverses the charging direction of the capacitor, resulting in continuous square wave generation at the output. D1, D2 control the symmetry and timing of the waveform by setting precise threshold voltages for switching.

Answer to Question 3 - The experiments constructed demonstrated the use of Op-Amps in waveform generation. The bipolar and uni-polar wave generator circuits showcased how feedback loops between a comparator and an integrator can produce continuous oscillations. The inclusion of diode clamping in the unipolar generator allowed control over the voltage range of the output. The astable multivibrator further illustrated the principle of self-triggering oscillation using positive feedback and RC timing. Across all circuits, the use of simple OP-Amp configurations enabled the generation of clear and predictable waveforms.